

Syllabus of B.Sc. I Theory Paper

Part A Introduction			
Program: Certificate		Class :B.Sc.	Year: I
		Session: 2025-26	
Subject: Botany			
1	Course Code		
2	Course Title	Fundamentals of Botany	
3	Course Type (Core)	Major -Paper I	
4	Pre-requisite (if any)	To study this course, a student must have had the subject Biology, Life science, Agriculture in class/12th/ certificate/diploma.	
5	Course Learning outcomes (CLO)	On completion of this course, the students will be able to: • Explore about the Indian knowledge System about plants. • Develop understanding about Plants • Identify the morphological and anatomical structures of plants on the basis of Characters • Will be able to understand the concepts and principles of physiological process of plants.	
6	Credit Value	4	
7	Total Marks	Max. Marks: 30+70	Min. Passing Marks:35
Part B- Content of the Course			
Total no of	Theory Lectures: 60 hours	Tutorials: 0	Credits: 4
Unit	Topics	No. of Lectures	
I	Plants in “Bhartiya Gyan Parampara” 1.1 Botany definition, scope and importance 1.2 History of Botany in India 1.3 Botanical concepts in Vedas, (Rig Veda, Atharvaveda) Upanishads, Ramayan and Puranas. 1.4 Contribution of Indian Rishis in botany (Parashar, Charak, Sushrut, Surpaal, Kautilya and others) 1.5 Plants in Indian Traditional food and Folk medicine <i>*Activity- listing plants in scriptures local traditions, Preparation of Charts /discussions on plants and Rishis, /field visit to identify plants.</i>	12	
II	Morphology-External Structure of Plants: 1.1 Basic morphological terms; Thallus and concept of plant body 1.2 Structure and Modifications of -Roots 1.3 Structure and Modifications of-Stem 1.4 Structure and Modifications of-Leaf . 1.5 Flower and types of InflorescencesandFruits and Seeds <i>* Field visit to study various types of leaves, flowers, inflorescence</i>	12	

III	Anatomy – Internal Structure of Plants: 1.1 Plant Cell- structure and theory 1.2 Tissues and types of Tissues 1.3 General Anatomy of -Root 1.4 General Anatomy of - Stem 1.5 General Anatomy of –Leaf <i>* Activity -preparation of charts and models</i>	12
IV	General idea of Plant Physiology: 1.1 Concepts of water as universal solvent, Imbibition, Diffusion, Osmosis and Water Potential 1.2 Transpiration and movement of water in Plants. 1.3 Transport of Solutes and ascent of sap 1.4 Assimilation of Energy- Photosynthesis 1.5 Release of Energy- Respiration <i>* Activity-Quiz competition on topic related to plant physiology</i>	12
V	Plant Reproduction, Growth and Propagation: 1.1 Plant propagation and types of propagules 1.2 Plant reproduction (Sexual and Asexual) 1.3 Plant growth and development (Role of Elicitors) 1.4 Concept of gametophyte and sporophyte 1.5 Alternation of Generation <i>* Activity- Field study/trail on germination of various seeds and propagation techniques.</i>	12

Keywords/Tags: Botany, Morphology, Plant Physiology, Plant reproductions

Part C-Learning Resources

Text Books, Reference Books, Other resources

Suggested Readings:

- Bhojwani, S.S. and Bhatnagar, S.P. (2011). The Embryology of Angiosperms, Vikas Publishing House. Delhi. 5th edition.
- Dickison, W.C. (2000). Integrative Plant Anatomy. Harcourt Academic Press, USA.
- Evert, R.F. (2006) Esau's Plant Anatomy: Meristems, Cells, and Tissues of the Plant Body: Their Structure, Function and Development. John Wiley and Sons, Inc.
- Fahn, A. (1974). Plant Anatomy. Pergamon Press, USA.
- Gangulee, H.C., Das, A.K., Dutta R.C. (1991) College Botany Vol -I New Central Book Agency, Kolkata
- Hopkins, W.G. and Huner, A. (2008). Introduction to Plant Physiology. John Wiley and Sons. U.S.A. 4th edition.
- Mauseth, J.D. (1988). Plant Anatomy. The Benjamins/Cummings Publisher, USA.
- Mishra, P.K. (2016) Botany in Vedas, Write and print Publications, New Delhi.
- Sharma, P.V. (1981) "Charaka Samhita" (Translation) Chaukhambha Orientalia, New Delhi
- Sircar, N. N. (1991) Vrakshayurveda of Parashar, (A Treatise on Plant Science) Sri Satgura Publication, New Delhi
- Srikantha Murthy, K.R. (1991) "Sushruta Samhita" – Translated: Chaukhambha Orientalia, New Delhi
- Taiz, L., Zeiger, E., Moller, I.M. and Murphy, A (2015). Plant Physiology and Development. Sinauer Associates Inc. USA. 6th edition.
- बीणासत्या (2022) आधारभूत वनस्पति शास्त्र - द्वितीय प्रश्नपत्र (प्रथम वर्ष)- मध्य प्रदेश हिंदी ग्रंथ अकादमी, भोपाल

2. Suggestive digital platforms web link

<https://highereducation.mp.gov.in/?page=2wxzkCTrFZmhC4di2oGkQw%3D%3D>

<https://iksindia.org/documentaries.php>

Suggested equivalent online courses:

<https://www.mooc.org>

<https://swayam.gov.in>

<https://nptel.ac.in>

Part D-Assessment and Evaluation

Suggested Continuous Evaluation Methods:

Maximum Marks: 100

Continuous Comprehensive Evaluation (CCE): 30 marks and University Exam (UE): 70 marks

Internal Assessment: Continuous Comprehensive Evaluation (CCE):30	Class Test Assignment/Presentation	30
External Assessment: University Exam Section: 70 Time: 03.00 Hours	Section(A): Very Short Answer Type Questions (50 Words Each) Section (B): Short answer type Questions (200 Words Each) Section (C): Long answer type Questions (500 Words Each)	Total 70

Any remarks/ suggestions:



Syllabus of Practical Paper

Part A Introduction			
Program: Certificate Course	Class: B.Sc.	Year: I	Session: 2025-26
Subject: Botany Practicals			
1	Course Code		
2	Course Title	Fundamentals of Botany -Practical	
3	Course Type (Core.)	Major core course Paper I- Practical	
4	Pre-requisite (if any)	To Study this course, a student must have studied the subject Biology/Life Sciences/Agriculture in class 12th.	
5	Course Learning Outcomes (CLO)	On completion of this course, learners will be able to: - Students will critically assess the contributions of ancient Indian scholars (Rishis) to the field of botany - Students will compare and contrast the anatomical structures of plants - Student will understand the functions of plants	
6	Credit Value	2 credits	
7	Total Marks 100	Max. Marks: 100	Min. Passing Marks:35
Part B- Content of the Course			
Total No. of hours of Practical: 30 hours			
Suggested list of Exercise:			
- To contextualize and list the plants from Ancient Indian texts. - Prepare a note on contributions of Rishi in botanical context, for example (Parashar/Charak/Sushrut/Surpaal/ and Kautilya - Documentation of traditional uses of plants or their parts in food and medicine. - Collect and categorize plants based on morphological features. (Leaves and Inflorescence) - Sketch and label structures of roots, stems, leaves, flowers, fruits, and seeds. - Prepare a thin section of an onion peel and stain it to observe cells. - Examine tissue types (parenchyma, collenchyma, sclerenchyma) in different plant sections. - Observe vascular bundles in roots, stems, and leaves. - Identify xylem, phloem, epidermis, and stomata. - Compare monocot and dicot anatomy. - Dissect typical flower and identify sepals, petals, stamens, and carpels			
Note: Site the sources of information collected.			
Remark:			
- Above practical's could be designed according to locally available plant materials. - Online available resources, charts models can also be used. - Field visits could be arranged based on the availability of the plant material.			
Keywords/Tags: Practical botany, Morphology, anatomy, Physiology,			
Part C-Learning Resources			
Text Books, Reference Books, Other resources			
Suggested Readings:			
Bendre, Ashok and Kumar, Ashok (1981) Practical Botany-2, Rastogi Publications, Meerut Bettany, G.T. (0) First lessons in Practical Botany Gyan Publishing House, Delhi			

Ragland, A. and Arumugam, N. (2020) Fundamentals of Plant Anatomy and Micro techniques, Saras Publications

Suggestive digital platforms web links: .

<https://highereducation.mp.gov.in/?page=2wxzkCTrFZmhC4di2oGkQw%3D%3D>

<https://iksindia.org/documentaries.php>

Suggested equivalent online courses:

<https://www.mooc.org>

<https://swayam.gov.in>

<https://nptel.ac.in>

Part D-Assessment and Evaluation

Suggested Continuous Evaluation Methods:

Part D- Scheme of the practical exam (External assessment)

Suggested Exercise:

1. Lab exercise based on external morphology/floral morphology	15
2. Lab exercise based on the anatomy of plant parts	15
3. Lab exercise based on plant physiology	10
4. Lab exercise based on plant propagation/ Documentation of Plants in Scriptures	10
5. Spotting (Spots 1 to 5)	20
6. Viva voce	10
7. Record and Sessional	10
8. Attendance	10
TOTAL=	100

Any remarks/ suggestions: